FACULTY OF COMPUTING

SEMSTER 2 2025/2026

SECI1143 PSDA

SECTION 02

ASSIGNMENT 1

LECTURER:DR.ROZILAWATI

GROUP 1:

STUDENT NAME	MATRIC NO
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Question 1.

a) Population > All students of the university who visited the library during the first week of the semester.

Sample > 60 randomly selected students who visited the library during the first week of the semester.

b) QUANTITATIVE

- Total number of books borrowed this semester.
- Time spent studying in the library per week (in hours)
- Temperature of the reading room when they visit (in celsius)

QUALITATIVE

- Student ID
- Major of study (e.g. Engineering, Arts, Business).
- Academic year (Year 1, Year 2, Year 3, Year 4).

c) DISCRETE

- Total number of books borrowed this semester.

CONTINUOUS

- Time spent studying in the library per week (in hours).
- Temperature of the reading room when they visit (in celsius).

d) Major of study = Nominal

Academic Year = Ordinal

Total number of books borrowed = Ratio.

Temperature of the reading room = Interval.

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Question 2

a)

Beverage	frequency	relative frequency	cumulative relative frequency
latte	45	0.38	0.38
expresso	30	0.25	0.63
cappucino	25	0.21	0.84
americano	20	0.17	1.01 ≈ 1.00
	total = 120		

relative frequency = frequency / total

rf latte = 45 / 120 = 0.375 ≈ 0.38

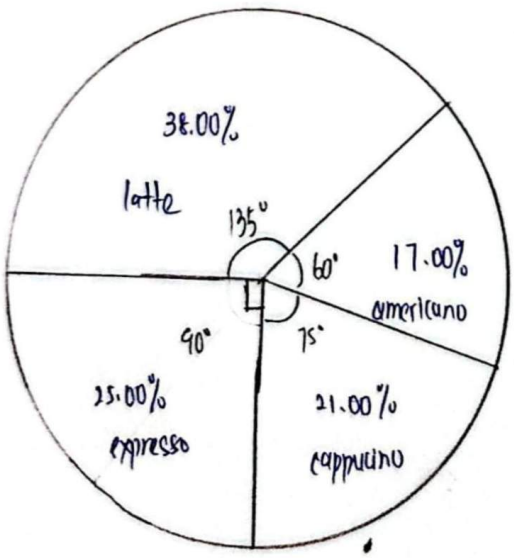
rf expresso = 30/120 = 0.25

rf cappucino = 25 / 120 = 0.208 ≈ 0.21

rf americano = 20 / 120 = 0.167 ≈ 0.17

b) in graph paper

c)



percentage :

latte = 0.38 × 100%
= 38.00%

expresso = 0.25 × 100%
= 25%

americano = 0.17 × 100%
= 17.00%

cappucino = 0.21 × 100%
= 21%

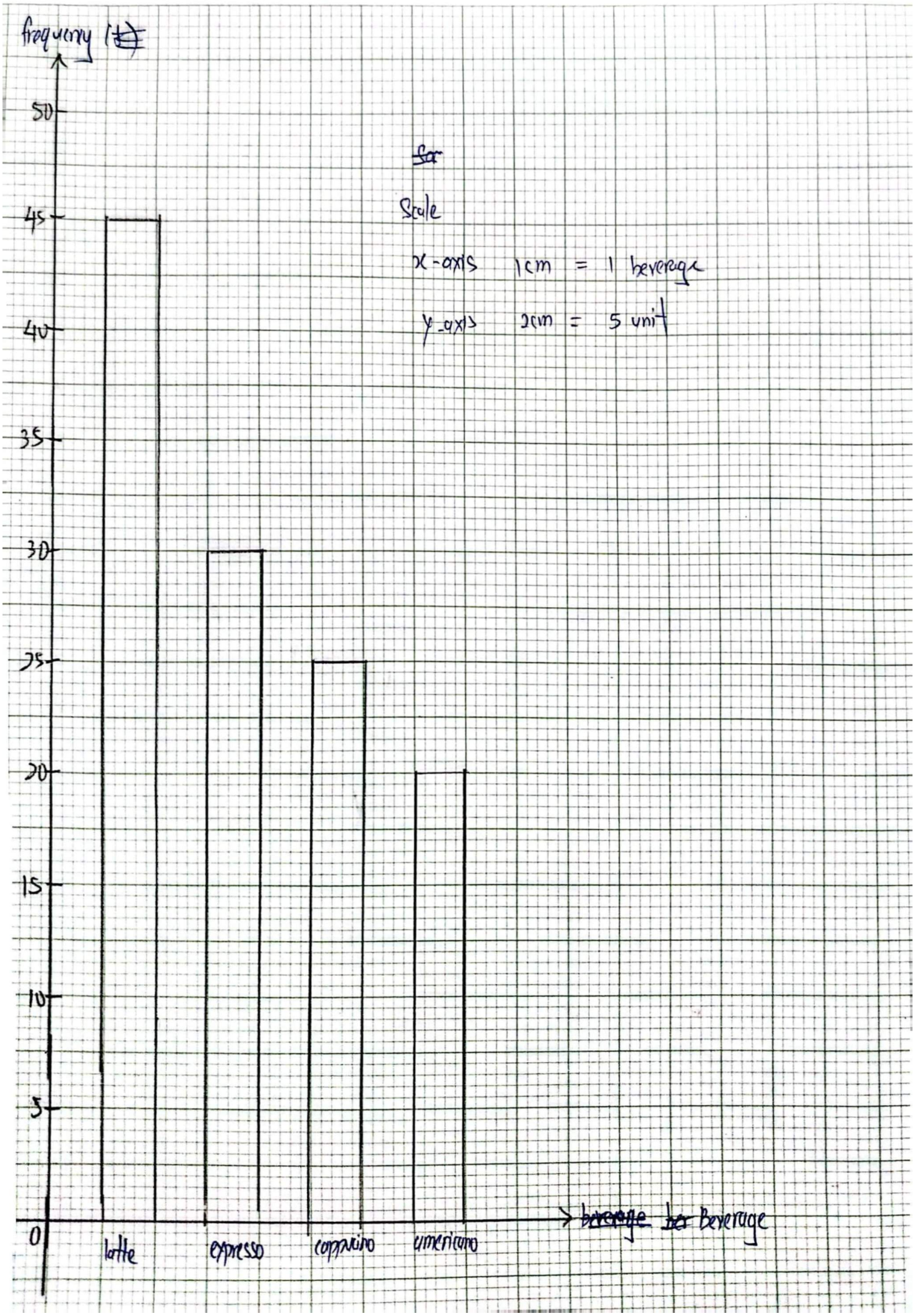
degree :

latte = 0.375 × 360°
= 135°

expresso = 0.25 × 360°
= 90°

americano = 0.167 × 360
= 60°

cappucino = 0.208 × 360
= 75°


frequency (f)
50
45
40
35
30
25
20
15
10
5
0
~~for~~
Scale
x-axis 1cm = 1 beverage
y-axis 2cm = 5 unit
latte
expresso
cappuccino
americano
~~beverage~~ ~~ber~~ Beverage

Question 3

(a) mean $= \dfrac{15 + 18 + 20 + 22 + 22 + 25 + 26 + 28 + 30 + 32 + 35 + 40 + 42 + 65 + 70 + 75}{16}$

$= \dfrac{565}{16}$

$= 35.31$ (2.dp)

median $= \dfrac{16}{2} = 8^{th}$

$\dfrac{8^{th} + 9^{th}}{2}$

$= \dfrac{28+30}{2}$

$= 29.00$

mode $= 22$

(b) $Q_1 = \dfrac{Y[i] + Y[i+1]}{2}$ $\quad i = \dfrac{25}{100} \times 16$

$= \dfrac{Y[4] + Y[5]}{2}$ $\quad = 4$ (integer)

$= \dfrac{22+22}{2}$

$= 22$

$Q_3 = \dfrac{Y[12] + Y[13]}{2}$ $\quad i = \dfrac{75}{100} \times 16$

$= 12$ (integer)

$= \dfrac{40+42}{2}$

$= 41$

iqr $= Q_3 - Q_1$

$= 41 - 22$

$= 19$

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(c) lower limit = $Q_1 - 1.5 \times iqr$
(mild boundary)

$= 22 - (1.5 \times 19)$

$= -6.50$

upper limit = $Q_3 + 1.5 \times iqr$
(mild boundary)

$= 41 + 1.5 \times 19$

$= 69.50$

Outliers (< -6.50) = None

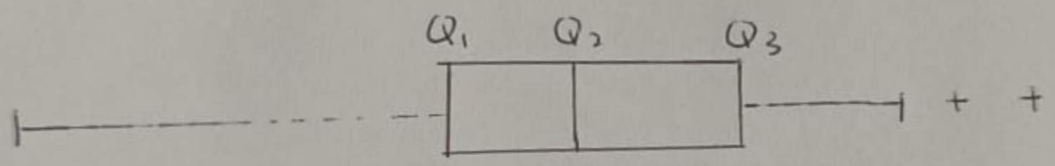
Outliers (> 69.50) = 70, 75

$\therefore$ 70, 75 are mild outliers

$1.5 \times iqr \rightarrow$ mild boundary

$3 \times iqr \rightarrow$ extreme boundary

upper limit (extreme) $41 + 3 \times 19 = 98$

$69.50 < 70, 75 < 98$

(d)